

Cherry Blossom Run - Population information

MATH 213

What denominator to use?

	F	M	Total Racers
10 mile	10446	6996	17442
5K	1821	698	2519

Group 1

Here is a table of information about the **population** of Cherry Blossom run participants.

	F	M
10 mile	10446	6996
5K	1821	698

Fill out the following table based on this information about the population.

Proportion of F in 10 mile	Proportion of F in 5K	Difference (10 mile – 5K)
0.60	0.72	-0.12

Are the numbers you computed above statistics or parameters? How can you tell?

Parameter, it's based on the population

Imagine taking a random 50-person sample from the 5K runners and a random 50-person sample from the 10-mile runners. Now imagine repeatedly taking such samples, computing the difference in proportions of females between the groups (10-mile – 5K), and plotting those differences on a dotplot. If we do this enough times, we will have simulated a **sampling distribution** for the difference in sample proportions.

Based on your experience with sampling distributions, what do you expect to see?

Center: -0.12

Shape: Normal

Spread: (Make a guess!) 95%: (-0.16, 0.02)

Here, we perform some computations that we will need later on. Show your work!

According to our standard-error formulas, what are the standard errors for the sample proportions of “F” in each race?

SE for 5K: 0.069

SE for 10 mile: 0.063

Group 2

Here is a table of information about the **population** of Cherry Blossom run participants.

	F	M
10 mile	10446	6996
5K	1821	698

Fill out the following table based on this information about the population.

Proportion of F in 10 mile	Proportion of F in 5K	Difference (10 mile – 5K)
0.5988992088	0.72290591504	-0.12400670624

Are the numbers you computed above statistics or parameters? How can you tell?
These numbers are parameters, since they refer to the entire population of Cherry Blossom run participants.

Imagine taking a random 50-person sample from the 5K runners and a random 50-person sample from the 10-mile runners. Now imagine repeatedly taking such samples, computing the difference in proportions of females between the groups (10-mile – 5K), and plotting those differences on a dotplot. If we do this enough times, we will have simulated a **sampling distribution** for the difference in sample proportions.

Based on your experience with sampling distributions, what do you expect to see?

Center: ~ -0.12

Shape: normal

Spread: (Make a guess!) ± 0.4

Here, we perform some computations that we will need later on. Show your work!

According to our standard-error formulas, what are the standard errors for the sample proportions of “F” in each race?

SE for 5K:

SE for 10 mile:

Group 3

Here is a table of information about the **population** of Cherry Blossom run participants.

	F	M
10 mile	10446	6996
5K	1821	698

Fill out the following table based on this information about the population.

Proportion of F in 10 mile	Proportion of F in 5K	Difference (10 mile – 5K)
17442 60%	2519 72.3%	$0.5989 - 0.7231 = -0.1242$ or -12.4%

Are the numbers you computed above statistics or parameters? How can you tell?

Parameter since the data comes from all Cherry Blossom run participants, not just a sample of runners.

Imagine taking a random 50-person sample from the 5K runners and a random 50-person sample from the 10-mile runners. Now imagine repeatedly taking such samples, computing the difference in proportions of females between the groups (10-mile – 5K), and plotting those differences on a dotplot. If we do this enough times, we will have simulated a **sampling distribution** for the difference in sample proportions.

Based on your experience with sampling distributions, what do you expect to see?

Center: Should be equal to the difference in population proportions

Shape: should be normal curve

Spread: (Make a guess!) 0.1

Here, we perform some computations that we will need later on. Show your work!

According to our standard-error formulas, what are the standard errors for the sample proportions of “F” in each race?

SE for 5K: Prop of F = $1821 / (1821 + 698) = 0.7231$
square root $(0.7231 * (1 - 0.7231)) / 50 = 0.0633$

SE for 10 mile: Proportion of F = $10446 / (10446 + 6996) = 0.5989$
square root $(0.5989 * (1 - 0.5989)) / 50$

Group 4

Here is a table of information about the **population** of Cherry Blossom run participants.

	F	M
10 mile	10446	6996
5K	1821	698

Fill out the following table based on this information about the population.

Proportion of F in 10 mile	Proportion of F in 5K	Difference (10 mile – 5K)
$10446/(10446+6996)$ =0.5988992=.60	$1821/(1821+698)$ =0.7229059=.72	-0.1240067 =-.12

Are the numbers you computed above statistics or parameters? How can you tell?

Parameters because population

Imagine taking a random 50-person sample from the 5K runners and a random 50-person sample from the 10-mile runners. Now imagine repeatedly taking such samples, computing the difference in proportions of females between the groups (10-mile – 5K), and plotting those differences on a dotplot. If we do this enough times, we will have simulated a **sampling distribution** for the difference in sample proportions.

Based on your experience with sampling distributions, what do you expect to see?

Center: **-.12**

Shape: **Standard bell shape**

Spread: **-.72 and .60**

Here, we perform some computations that we will need later on. Show your work!

According to our standard-error formulas, what are the standard errors for the sample proportions of “F” in each race?

SE for 10 mile: **.069**

SE for 5K: **.063**

Group 5

Here is a table of information about the **population** of Cherry Blossom run participants.

	F	M
10 mile	10446	6996
5K	1821	698

Fill out the following table based on this information about the population.

Proportion of F in 10 mile	Proportion of F in 5K	Difference (10 mile – 5K)
0.598899208806	0.722905915046	-0.12400670624

Are the numbers you computed above statistics or parameters? How can you tell?
Parameters because they apply to the population

Imagine taking a random 50-person sample from the 5K runners and a random 50-person sample from the 10-mile runners. Now imagine repeatedly taking such samples, computing the difference in proportions of females between the groups (10-mile – 5K), and plotting those differences on a dotplot. If we do this enough times, we will have simulated a **sampling distribution** for the difference in sample proportions.

Based on your experience with sampling distributions, what do you expect to see?

Center: 0

Shape: bell curve

Spread: (Make a guess!) -.23- .23

Here, we perform some computations that we will need later on. Show your work!

According to our standard-error formulas, what are the standard errors for the sample proportions of “F” in each race?

SE for 5K: $\sqrt{0.722905915046(1-0.722905915046)/2519}= 0.00891744486725$

SE for 10 mile: $\sqrt{\frac{0.598899208806}{17442}(1-0.598899208806)}= 0.00371112418992$

Group 6

Here is a table of information about the **population** of Cherry Blossom run participants.

	F	M
10 mile	10446	6996
5K	1821	698

Fill out the following table based on this information about the population.

Proportion of F in 10 mile	Proportion of F in 5K	Difference (10 mile – 5K)
0.5988992088	0.72290591504	-0.12400670624

Are the numbers you computed above statistics or parameters? How can you tell?

They are parameters because they describe the population of Cherry Blossom run participants.

Imagine taking a random 50-person sample from the 5K runners and a random 50-person sample from the 10-mile runners. Now imagine repeatedly taking such samples, computing the difference in proportions of females between the groups (10-mile – 5K), and plotting those differences on a dotplot. If we do this enough times, we will have simulated a **sampling distribution** for the difference in sample proportions.

Based on your experience with sampling distributions, what do you expect to see?

Center: -0.124

Shape: Bell shaped curve

Spread: (Make a guess!) 95%

Here, we perform some computations that we will need later on. Show your work!

According to our standard-error formulas, what are the standard errors for the sample proportions of “F” in each race?

SE for 5K:

SE for 10 mile:

Group 7

Here is a table of information about the **population** of Cherry Blossom run participants.

	F	M
10 mile	10446	6996
5K	1821	698

Fill out the following table based on this information about the population.

Proportion of F in 10 mile	Proportion of F in 5K	Difference (10 mile – 5K)
0.599	0.723	-0.124

Are the numbers you computed above statistics or parameters? How can you tell?
They are parameters because its about the population

Imagine taking a random 50-person sample from the 5K runners and a random 50-person sample from the 10-mile runners. Now imagine repeatedly taking such samples, computing the difference in proportions of females between the groups (10-mile – 5K), and plotting those differences on a dotplot. If we do this enough times, we will have simulated a **sampling distribution** for the difference in sample proportions.

Based on your experience with sampling distributions, what do you expect to see?

Center: -0.124

Shape: Normal

Spread: (Make a guess!) $\pm 3(0.124)$

Here, we perform some computations that we will need later on. Show your work!

According to our standard-error formulas, what are the standard errors for the sample proportions of “F” in each race?

SE for 5K: $\sqrt{p(1-p)/n}$ $\sqrt{0.599(1-0.599)/50}$ SE = 0.0062

SE for 10 mile: $\sqrt{p(1-p)/n}$ $\sqrt{0.723(1-0.723)/50}$ SE = 0.0047

Group 8

Here is a table of information about the **population** of Cherry Blossom run participants.

	F	M
10 mile	10446	6996
5K	1821	698

Fill out the following table based on this information about the population.

Proportion of F in 10 mile	Proportion of F in 5K	Difference (10 mile – 5K)
$10446 / (10446 + 6996)$ = 0.5988992 (59.9%)	$1821 / (1821 + 698)$ = 0.7229059 (72.3%)	-0.124

Are the numbers you computed above statistics or parameters? How can you tell?

Parameters

Imagine taking a random 50-person sample from the 5K runners and a random 50-person sample from the 10-mile runners. Now imagine repeatedly taking such samples, computing the difference in proportions of females between the groups (10-mile – 5K), and plotting those differences on a dotplot. If we do this enough times, we will have simulated a **sampling distribution** for the difference in sample proportions.

Based on your experience with sampling distributions, what do you expect to see?

Center: Population difference in proportion (-0.124)

Shape: Normal distribution / bell-shaped

Spread: (Make a guess!) +/- 0.1

Here, we perform some computations that we will need later on. Show your work!

According to our standard-error formulas, what are the standard errors for the sample proportions of “F” in each race?

SE for 5K: $\sqrt{0.7229 \cdot (1 - 0.7229) / 50} = 0.06329543$

SE for 10 mile: $\sqrt{0.5988 \cdot (1 - 0.5988) / 50} = 0.0693165$

Group 9

Here is a table of information about the **population** of Cherry Blossom run participants.

	F	M
10 mile	10446	6996
5K	1821	698

Fill out the following table based on this information about the population.

Proportion of F in 10 mile	Proportion of F in 5K	Difference (10 mile – 5K)
$10446/17442 = .60\%$	0.72	-0.12

Are the numbers you computed above statistics or parameters? How can you tell?

Parameter because its a population of all women runners

Imagine taking a random 50-person sample from the 5K runners and a random 50-person sample from the 10-mile runners. Now imagine repeatedly taking such samples, computing the difference in proportions of females between the groups (10-mile – 5K), and plotting those differences on a dotplot. If we do this enough times, we will have simulated a **sampling distribution** for the difference in sample proportions.

Based on your experience with sampling distributions, what do you expect to see?

Center: - 0.120

Shape: Normal bell shape

Spread: (Make a guess!)

Here, we perform some computations that we will need later on. Show your work!

According to our standard-error formulas, what are the standard errors for the sample proportions of “F” in each race?

SE for 5K: $.72/\sqrt{1-.72} = .0169$

SE for 10 mile:

Group 10

Here is a table of information about the **population** of Cherry Blossom run participants.

	F	M
10 mile	10446	6996
5K	1821	698

Fill out the following table based on this information about the population.

Proportion of F in 10 mile	Proportion of F in 5K	Difference (10 mile – 5K)
$10446 / (10446 + 6996) = 0.5988$	$1821 / (1821 + 698) = 0.72$	-0.124

Are the numbers you computed above statistics or parameters? How can you tell?
They are parameters because they describe the runners in the cherry blossom run.

Imagine taking a random 50-person sample from the 5K runners and a random 50-person sample from the 10-mile runners. Now imagine repeatedly taking such samples, computing the difference in proportions of females between the groups (10-mile – 5K), and plotting those differences on a dotplot. If we do this enough times, we will have simulated a **sampling distribution** for the difference in sample proportions.

Based on your experience with sampling distributions, what do you expect to see?

Center: -.124

Shape: normal / bell shaped

Spread: (Make a guess!) ± 0.1

Here, we perform some computations that we will need later on. Show your work!

According to our standard-error formulas, what are the standard errors for the sample proportions of “F” in each race?

SE for 5K: $SE = \sqrt{p(1-p)/n} = \sqrt{0.72(1-0.72)/50} = 0.063$ (prob a little off because I rounded the 0.72)

SE for 10 mile: $\sqrt{0.5988(1-0.5988)/50} = 0.069$